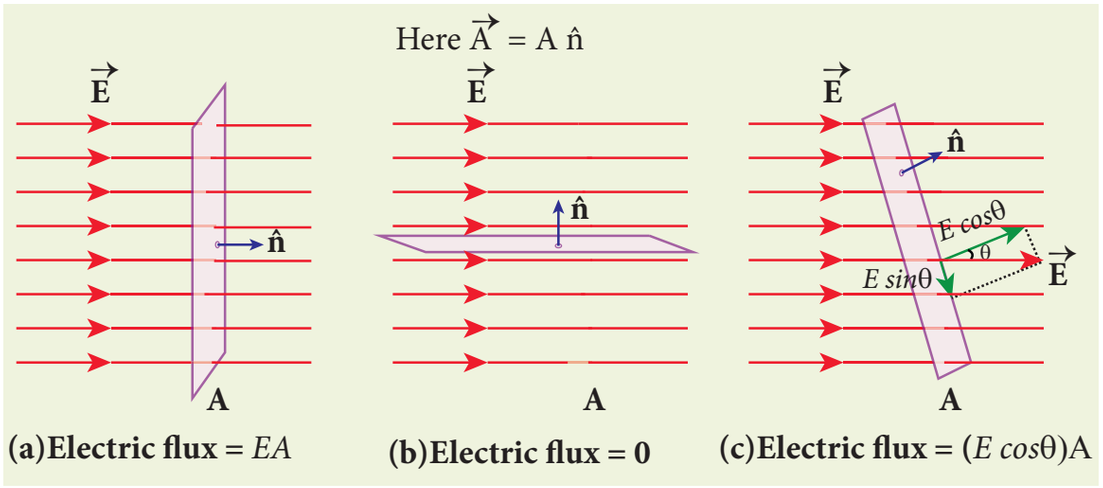


Figure 1.33 The electric flux for Uniform electric field

$$\Phi_E = \vec{E}\cdot\vec{A} = EA\cos\theta \qquad (1.55)$$

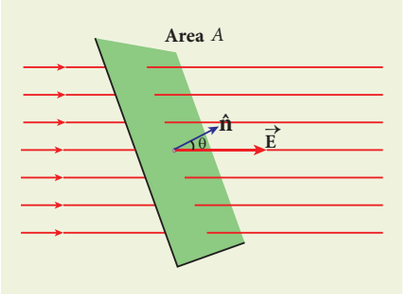
Here, note that $\vec{A}$ is the area vector $\vec{A} = A\hat{n}$. Its magnitude is simply the area A and the direction is along the unit vector $\hat{n}$ perpendicular to the area as shown in Figure 1.33. Using this definition for flux, $\Phi_E = \vec{E}\cdot\vec{A}$, equations (1.53) and (1.54) can be obtained as special cases.

In Figure 1.33 (a), $\theta = 0^o$ so $\Phi_E = \vec{E}\cdot\vec{A} = EA$

In Figure 1.33 (b), $\theta = 90^o$ so $\Phi_E = \vec{E}\cdot\vec{A} = 0$

EXAMPLE 1.17

Calculate the electric flux through the rectangle of sides 5 cm and 10 cm kept in the region of a uniform electric field 100 NC^{-1}. The angle θ is 60°. Suppose θ becomes zero, what is the electric flux?



Solution

The electric flux

$$\Phi_E = \vec{E}\cdot\vec{A} = EA\cos\theta = 100\times5\times10\times10^{-4}\times\cos60^\circ$$

$$\Rightarrow \qquad \Phi_E = 0.25\,N.m^2C^{-1}$$

For θ = 0°,

$$\Phi_E = \vec{E}\cdot\vec{A} = EA = 100\times5\times10\times10^{-4} = 0.5\,N.m^2C^{-1}$$

Electric flux in a non uniform electric field and an arbitrarily shaped area

Suppose the electric field is not uniform and the area A is not flat (Figure 1.34), then the entire area is divided into *n* small area segments $\Delta\vec{A}_1, \Delta\vec{A}_2, \Delta\vec{A}_3 \ldots\ldots \Delta\vec{A}_n$ such that each area element is almost flat and the electric field over each area element is considered to be uniform.

The electric flux for the entire area A is approximately written as

$$\Phi_E = \vec{E}_1\cdot\Delta\vec{A}_1 + \vec{E}_2\cdot\Delta\vec{A}_2 + \vec{E}_3\cdot\Delta\vec{A}_3 \ldots\ldots \vec{E}_n\cdot\Delta\vec{A}_n$$
$$= \sum_{i=1}^{n}\vec{E}_i\cdot\Delta\vec{A}_i \qquad (1.56)$$

UNIT-1(XII-Physics_Vol-1).indd 39 04-03-2019 10:26:00